CTCF & Enhancer Data Processing – Reproducibility Notes

1. Directory Structure

```
Improved Project Alpha/
■■■ data/
    ■■■ raw/
        ■■■ dog/
        ■   ■■■ enhancers/
        ■       ■■■ GSE203104_k27ac.narrowpeak.tar.gz
        ■       ■■■ ChIP_CL_k27ac_optimal.narrowPeak
        ■       ■■■ ChIP_CO_k27ac_optimal.narrowPeak
        ■       ■■■ ... (other tissue-specific enhancer files)
        ■■■ mouse/
            ■■■ ctcf/
                ■■■ brain/
                    ■■■ *.bigBed
                    ■■■ *.bed.gz (converted from bigBed)
                    ■■■ mouse_brain_CTCF_union.bed.gz
```

2. Enhancer Data (Dog)

```
- Downloaded from GEO (GSE203104):
  curl -L -o GSE203104_k27ac.narrowpeak.tar.gz     "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE203nnn/GS
  tar -xvf GSE203104_k27ac.narrowpeak.tar.gz

- Resulting files:
  - 11 tissue-specific *_k27ac_optimal.narrowPeak enhancer sets.
  - Located in: data/raw/dog/enhancers/
```

3. CTCF Data (Mouse, Brain)

```
Step A: Conversion from .bigBed → .bed.gz
for f in data/raw/mouse/ctcf/brain/*.bigBed; do
    base=$(basename "$f" .bigBed)
    out="data/raw/mouse/ctcf/brain/${base}.bed.gz"
    if [ ! -f "$out" ]; then
        echo "Converting $f → $out"
        bigBedToBed "$f" stdout | gzip -c > "$out"
    fi
done

Step B: Merge into Union Set
DIR="data/raw/mouse/ctcf/brain"
gzcat "$DIR"/*.bed.gz | awk 'BEGIN{OFS="\t"}{print $1,$2,$3}'   | LC_ALL=C sort -k1,1 -k2,2n   | bedtool

gzip -f "$DIR/mouse_brain_CTCF_union.bed"

Output:
- data/raw/mouse/ctcf/brain/mouse_brain_CTCF_union.bed.gz
- Size: ~11 MB
- Format: BED intervals (chr start end)
```

4. What’s Missing

```
- Dog CTCF data
  - Not yet downloaded.
```

```
- Need to identify and fetch dog CTCF ChIP-seq datasets (likely from GEO or ArrayExpress).
- Once obtained, repeat the same pipeline:
  1. Download .bigBed or .bed files.
  2. Convert to .bed.gz if needed.
  3. Merge into dog_<tissue>_CTCF_union.bed.gz.
```

5. Next Steps

```
1. Locate available dog CTCF ChIP-seq datasets.
2. Mirror the mouse pipeline (conversion + merging).
3. Document download sources (GEO accession IDs, ENCODE if available).
```